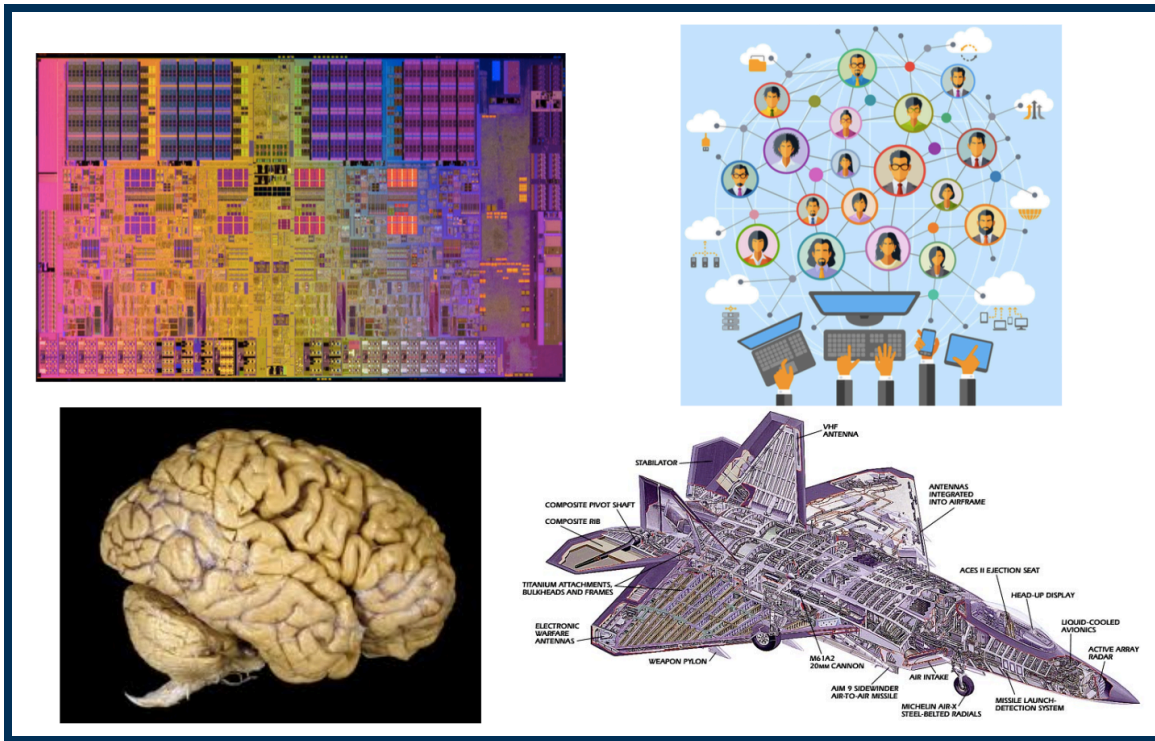


L1: Complex Systems



The image above shows a microprocessor, a human brain, an online social network, and a fighter jet. On the surface, you may think that these systems have nothing in common!

However, they all have some common fundamental properties:

1. Each of them consists of many autonomous parts, or modules – in the same way, that a large puzzle consists of many little pieces. The microprocessor, for example, consists of mostly transistors and interconnects. The brain consists of various cell types, including excitatory neurons, inhibitory neurons, glial cells, etc.
2. The parts of each system are not connected randomly or in any other trivial manner – on the contrary, the system only works if the connections between the parts are highly specific (for example, we would not expect an electronic device to work if its transistors were randomly connected). These interconnections between the system components define the architecture of the system – or in other words, the network representation of the system.
3. The interactions between connected parts are also non-trivial. A trivial interaction would be, mathematically, a linear relation between the activity of two parts. On the

contrary, as we will discuss later in the course, in all interesting systems at least, these interactions are non-linear.

To summarize, Complex Systems have:

- Many and heterogeneous components
- Components that interact with each other through a (*non-trivial*) network
- Non-linear interactions between components

Next, we'll discuss Trivial Networks versus Complex Networks.